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The gauge theory of dislocations: A nonuniformly moving screw dislocation

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ABSTRACT

We investigate the nonuniform motion of a straight screw dislocation in infinite media in the framework of the translational gauge theory of dislocations. The equations of motion are derived for an arbitrarily moving screw dislocation. The fields of the elastic velocity, elastic distortion, dislocation density and dislocation current surrounding the arbitrarily moving screw dislocation are derived explicitly in the form of integral representations. We calculate the radiation fields and the fields depending on the dislocation velocities.

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1. Introduction

For many years, the investigation of the nonuniform motion of dislocations has attracted the interest of researchers in different fields such as physics, material science, continuum mechanics, seismology and earthquake engineering. Usually, the motion of dislocations is investigated in the framework of incompatible elastodynamics where the dislocation density tensor and the dislocation current tensor are given as source terms of the elastic fields (see, e.g., Mura [1,2], Lardner [3], Günther [4]). The behaviour of the motion of a dislocation is somehow particular, because at any time the fields are determined not only by the instantaneous values of the velocity (or higher derivatives of the position with respect to time), but also by the values of these quantities in the past [5,6]. As Eshelby [5] succinctly put it: 'The dislocation is haunted by its past'. This fact is based on the property that Huygens' principle is not valid in two dimensions (see, e.g., [7]). Solutions for nonuniformly moving screw dislocations have been given by Nabarro [8], Eshelby [5,6], Kiusalaas and Mura [9], and Markenscoff [10]. Results for the nonuniform motion of a gliding edge dislocation have been given by Kiusalaas and Mura [11], Markenscoff and Clifton [12], and Brock [13].

Immediately, the question comes up, what is the suitable theory for an improved approach of a continuum theory of dislocation dynamics. Generalized theories of elasticity are, for instance, nonlocal elasticity [14,15], gradient elasticity [16,17] and dislocation gauge theory [18–21]. However, in the dynamical version of nonlocal elasticity it is not obvious which dynamical kernel should be used. In the dynamical extension of gradient elasticity [22,23] the choice of 'dynamical' gradients and length scales is based on ad-hoc assumptions. On the other hand, dislocation gauge theory is not based on such assumptions. The choice of the dynamical state variables is given by a canonical field theoretical framework [21]. For this reason, dislocation gauge theory seems to be the appropriate theory and it will be used in this Letter.

Thus, a very promising and straightforward candidate for an improved dynamical approach of dislocations including scale effects is the translational gauge theory of dislocations [18,19,21,24]. In the gauge theory of dislocations, dislocations arise naturally as a consequence of broken translational symmetry and their existence is not required to be postulated a priori. Moreover, such a theory uses the field theoretical framework which is well accepted in theoretical physics. From the point of view of theoretical physics, the gauge field approach of dislocations [18,19,21,24] removes the singularity at the core of a moving dislocation at subsonic as well as at supersonic speed [25], the gauge theoretical approach in essence being reminiscent of nonlocality. Up to now, no solution for a nonuniformly moving dislocation has been given in the dislocation gauge theory or another generalized theory of elasticity. The aim of this Letter is the examination of a nonuniformly moving screw dislocation, for the first time, in the framework of dislocation gauge theory. We will calculate the retarded

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expressions of the elastic fields as well as of the dislocation density and dislocation current tensors. This Letter provides new insights on the change of the dislocation core structure while the dislocation is moving at sound wave speed.

2. Gauge theory of dislocations

In this section, we briefly review the gauge theory of dislocations in the form given by Lazar and Anastassiadis [21], and Lazar [25].

In dislocation dynamics, the following state quantities of dislocations are of importance[1]

$$T_{ijk} = \beta_{ik,j} - \beta_{ij,k}, \qquad I_{ij} = -v_{i,j} + \dot{\beta}_{ij}, \tag{1}$$

which are called the dislocation density tensor and the dislocation current tensor, respectively. They are kinematical quantities of dislocations and they are given in terms of the incompatible elastic distortion tensor β_{ij} and incompatible physical velocity of the material continuum v_i. Their dimensions are: $[\beta_{ij}] = 1$, $[v_i] = \text{length/time}$, $[T_{ijk}] = 1/\text{length}$ and $[I_{ij}] = 1/\text{time}$. The dislocation density and the dislocation current tensors fulfill the translational Bianchi identities

$$\epsilon_{jkl} T_{ijk,l} = 0, \qquad \dot{T}_{ijk} + I_{ij,k} - I_{ik,j} = 0. \tag{2}$$

In the dynamical translation gauge theory of dislocations, the Lagrangian density is of the bilinear form

$$\mathcal{L} = T - W = \frac{1}{2} p_i v_i + \frac{1}{2} D_{ij} I_{ij} - \frac{1}{2} \sigma_{ij} \beta_{ij} - \frac{1}{4} H_{ijk} T_{ijk}. \tag{3}$$

The canonical conjugate quantities (response quantities) are defined by

$$p_i := \frac{\partial \mathcal{L}}{\partial v_i}, \qquad \sigma_{ij} := -\frac{\partial \mathcal{L}}{\partial \beta_{ij}}, \qquad D_{ij} := \frac{\partial \mathcal{L}}{\partial I_{ij}}, \qquad H_{ijk} := -2 \frac{\partial \mathcal{L}}{\partial T_{ijk}}, \tag{4}$$

where p_i, σ_{ij}, D_{ij}, and H_{ijk} are the linear momentum vector, the force stress tensor, the dislocation momentum flux tensor, and the pseudomoment stress tensor,[2] respectively. They have the dimensions: $[p_i] = \text{momentum/(length)}^3 \overset{\text{SI}}{=} \text{N s/m}^3$, $[D_{ij}] = \text{momentum/(length)}^2 \overset{\text{SI}}{=} \text{N s/m}^2$, $[\sigma_{ij}] = \text{force/(length)}^2 \overset{\text{SI}}{=} \text{Pa}$ and $[H_{ijk}] = \text{force/length} \overset{\text{SI}}{=} \text{N/m}$.

The Euler–Lagrange equations derived from the total Lagrangian density $\mathcal{L} = \mathcal{L}(v_i, \beta_{ij}, I_{ij}, T_{ijk})$ are given by

$$E_i^{\boldsymbol{v}}(\mathcal{L}) = \partial_t \frac{\partial \mathcal{L}}{\partial \dot{v}_i} + \partial_j \frac{\partial \mathcal{L}}{\partial v_{i,j}} - \frac{\partial \mathcal{L}}{\partial v_i} = 0, \tag{5}$$

$$E_{ij}^{\boldsymbol{\beta}}(\mathcal{L}) = \partial_t \frac{\partial \mathcal{L}}{\partial \dot{\beta}_{ij}} + \partial_k \frac{\partial \mathcal{L}}{\partial \beta_{ij,k}} - \frac{\partial \mathcal{L}}{\partial \beta_{ij}} = 0. \tag{6}$$

We add to $\mathcal{L}$ a so-called null Lagrangian, $\mathcal{L}_N = \sigma^0_{ij} \beta_{ij} - p^0_i v_i$, with the 'background' stress σ^0_{ij} and the 'background' momentum p^0_i as external source fields, which satisfy: $\dot{p}^0_i - \sigma^0_{ij,j} = 0$. In terms of the canonical conjugate quantities (4), Eqs. (5) and (6) take the form

$$D_{ij,j} + p_i = p^0_i \quad \text{(momentum balance of dislocations)}, \tag{7}$$

$$\dot{D}_{ij} + H_{ijk,k} + \sigma_{ij} = \sigma^0_{ij} \quad \text{(stress balance of dislocations)}. \tag{8}$$

Eqs. (7) and (8) represent the dynamical equations for the balance of dislocations. Eq. (7) represents the momentum balance law of dislocations, where the physical momentum is the source of the dislocation momentum flux. Eq. (8) represents the stress balance of dislocations. The force stress and the time derivative of the dislocation momentum flux are the sources of the pseudomoment stress. It can be seen in Eqs. (7) and (8) that the sources of the dislocation momentum flux and the pseudomoment stress tensors are the so-called effective momentum vector $(p_i - p^0_i)$ and the effective force stress tensor $(\sigma_{ij} - \sigma^0_{ij})$ (see, e.g., [19]). These effective fields, which are the difference between the physical field and the background field, drive the dislocation fields.

The conservation of linear momentum appears as an integrability condition from (7) and (8). It reads

$$\dot{p}_i - \sigma_{ij,j} = 0 \quad \text{(force balance of elasticity)}, \tag{9}$$

where the time-derivative of the physical momentum vector is the source of the force stress tensor.

The linear, isotropic constitutive relations for the momentum, the force stress, the dislocation momentum flux and the pseudomoment stress are

$$p_i = \rho v_i, \tag{10}$$

$$\sigma_{ij} = \lambda \delta_{ij} \beta_{kk} + \mu(\beta_{ij} + \beta_{ji}) + \gamma(\beta_{ij} - \beta_{ji}), \tag{11}$$

$$D_{ij} = d_1 \delta_{ij} I_{kk} + d_2(I_{ij} + I_{ji}) + d_3(I_{ij} - I_{ji}), \tag{12}$$

$$H_{ijk} = c_1 T_{ijk} + c_2(T_{jki} - T_{kji}) + c_3(\delta_{ij} T_{llk} - \delta_{ik} T_{llj}), \tag{13}$$

[1] We use the usual notations: $\beta_{ij,k} := \partial_k \beta_{ij}$ and $\dot{\beta}_{ij} := \partial_t \beta_{ij}$.

[2] The moment stress tensor $\tau_{ijk} = -\tau_{jik}$ can be obtained from the pseudomoment stress tensor: $\tau_{ijk} = -H_{[ij]k}$ (see [24,26]).

where ρ is the mass density. Here μ, λ, γ are the elastic stiffness parameters, c_1, c_2, c_3 denote the resistivity parameters associated with dislocations and d_1, d_2, d_3 are the inertia terms associated with dislocation currents. The nine material parameters have the dimensions: $[\mu, \lambda, \gamma] = \text{force}/(\text{length})^2 \overset{\text{SI}}{=} \text{Pa}$, $[c_1, c_2, c_3] = \text{force} \overset{\text{SI}}{=} \text{N}$ and $[d_1, d_2, d_3] = \text{mass}/\text{length} \overset{\text{SI}}{=} \text{kg/m}$.

The requirement of non-negativity of the energy (material stability) $E = T + W \geqslant 0$ leads to the conditions of semi-positiveness of the constitutive moduli. Particularly, the constitutive moduli have to fulfill the following conditions [21]

$$\begin{array}{lll} \rho \geqslant 0, & & \\ \mu \geqslant 0, & \gamma \geqslant 0, & 3\lambda + 2\mu \geqslant 0, \\ d_2 \geqslant 0, & d_3 \geqslant 0, & 3d_1 + 2d_2 \geqslant 0, \\ c_1 - c_2 \geqslant 0, & c_1 + 2c_2 \geqslant 0, & c_1 - c_2 + 2c_3 \geqslant 0. \end{array} \tag{14}$$

If we substitute the constitutive equations (12) and (13) in Eqs. (7) and (8) and use the definitions (1), we find

$$d_1(\dot{\beta}_{jj,i} - v_{j,ji}) + (d_2 + d_3)(\dot{\beta}_{ij,j} - v_{i,jj}) + (d_2 - d_3)(\dot{\beta}_{ji,j} - v_{j,ji}) + p_i = p_i^0, \tag{15}$$

$$\begin{aligned} &d_1\delta_{ij}(\ddot{\beta}_{kk} - \dot{v}_{k,k}) + (d_2 + d_3)(\ddot{\beta}_{ij} - \dot{v}_{i,j}) + (d_2 - d_3)(\ddot{\beta}_{ji} - \dot{v}_{j,i}) + c_1(\beta_{ik,jk} - \beta_{ij,kk}) + c_2(\beta_{ji,kk} - \beta_{jk,ik} + \beta_{kj,ik} - \beta_{ki,jk}) \\ &\quad + c_3\big[\delta_{ij}(\beta_{lk,lk} - \beta_{ll,kk}) + (\beta_{kk,ji} - \beta_{kj,ki})\big] + \sigma_{ij} = \sigma_{ij}^0, \end{aligned} \tag{16}$$

which are a coupled system of partial differential equations for the fields $\boldsymbol{v}$ and $\boldsymbol{\beta}$.

3. Equations of motion of a screw dislocation

We now derive the equations of motion for an arbitrarily moving screw dislocation. We consider an infinitely long screw dislocation parallel to the z-axis and traveling in the xy-plane. The symmetry of such a straight screw dislocation leaves only the following non-vanishing components of the physical velocity vector and elastic distortion tensor (see, e.g., [3,4]): v_z, β_{zx}, β_{zy}, and for the dislocation density and dislocation current tensors: T_{zxy}, I_{zx}, I_{zy}. The equations of motion of a moving screw dislocation read

$$(d_2 + d_3)(\dot{\beta}_{zx,x} + \dot{\beta}_{zy,y} - \Delta v_z) + \rho v_z = \rho v_z^0, \tag{17}$$

$$(d_2 + d_3)(\ddot{\beta}_{zx} - \dot{v}_{z,x}) + c_1(\beta_{zy,xy} - \beta_{zx,yy}) + (\mu + \gamma)\beta_{zx} = (\mu + \gamma)\beta_{zx}^0, \tag{18}$$

$$(d_2 - d_3)(\ddot{\beta}_{zx} - \dot{v}_{z,x}) + c_2(\beta_{zx,yy} - \beta_{zy,xy}) + (\mu - \gamma)\beta_{zx} = (\mu - \gamma)\beta_{zx}^0, \tag{19}$$

$$(d_2 + d_3)(\ddot{\beta}_{zy} - \dot{v}_{z,y}) + c_1(\beta_{zx,xy} - \beta_{zy,xx}) + (\mu + \gamma)\beta_{zy} = (\mu + \gamma)\beta_{zy}^0, \tag{20}$$

$$(d_2 - d_3)(\ddot{\beta}_{zy} - \dot{v}_{z,y}) + c_2(\beta_{zy,xx} - \beta_{zx,xy}) + (\mu - \gamma)\beta_{zy} = (\mu - \gamma)\beta_{zy}^0, \tag{21}$$

where $\Delta = \partial_{xx} + \partial_{yy}$. In addition, the condition (9) reads now

$$(\mu + \gamma)(\beta_{zx,x} + \beta_{zy,y}) = \rho \dot{v}_z. \tag{22}$$

From Eqs. (18)–(21) we obtain the following relations

$$\frac{c_1}{\mu + \gamma} = -\frac{c_2}{\mu - \gamma}, \tag{23}$$

$$\frac{d_2 + d_3}{\mu + \gamma} = \frac{d_2 - d_3}{\mu - \gamma} \tag{24}$$

and we may introduce the quantities

$$\ell_1^2 = \frac{c_1}{\mu + \gamma}, \tag{25}$$

$$L_1^2 = \frac{d_2 + d_3}{\rho}, \tag{26}$$

$$\tau_1^2 = \frac{d_2 + d_3}{\mu + \gamma}. \tag{27}$$

Here ℓ_1 and L_1 are the 'static' and 'dynamic' characteristic length scales and τ_1 is the characteristic time scale of the anti-plane strain problem. Due to the conditions (14), they are non-negative: $\ell_1 \geqslant 0$, $L_1 \geqslant 0$, $\tau_1 \geqslant 0$. The velocity of elastic shear waves is defined in terms of the 'dynamic' length scale L_1 and the time scale τ_1:

$$c_{\mathrm{T}}^2 = \frac{L_1^2}{\tau_1^2} = \frac{\mu + \gamma}{\rho}. \tag{28}$$

In a similar way, we introduce the following transversal gauge-theoretical velocity defined in terms of ℓ_1 and τ_1:

$$a_{\mathrm{T}}^2 = \frac{\ell_1^2}{\tau_1^2} = \frac{c_1}{d_2 + d_3}, \tag{29}$$

and we find the relation

$$\frac{\ell_1^2}{L_1^2} = \frac{a_{\mathrm{T}}^2}{c_{\mathrm{T}}^2}. \tag{30}$$

Applying Eq. (22), the equations of motion (17)–(21) can be given in the form

$$\tau_1^2 \ddot{v}_z - L_1^2 \Delta v_z + v_z = v_z^0, \tag{31}$$

$$\tau_1^2 \ddot{\beta}_{zx} - \ell_1^2 \Delta \beta_{zx} - \tau_1^2 \left(1 - \frac{\ell_1^2}{L_1^2}\right) \dot{v}_{z,x} + \beta_{zx} = \beta_{zx}^0, \tag{32}$$

$$\tau_1^2 \ddot{\beta}_{zy} - \ell_1^2 \Delta \beta_{zy} - \tau_1^2 \left(1 - \frac{\ell_1^2}{L_1^2}\right) \dot{v}_{z,y} + \beta_{zy} = \beta_{zy}^0. \tag{33}$$

If we assume that the following relation is valid (see also [25])

$$L_1 = \ell_1, \tag{34}$$

then the field equations (31)–(33) decouple to Klein–Gordon equations. Eqs. (30) and (34) give the relation $a_{\mathrm{T}} = c_{\mathrm{T}}$, that means that we have only one characteristic velocity c_{T} under this assumption. Particularly, the elastic fields fulfill the uncoupled Klein–Gordon equations

$$\left[1 + \ell_1^2 \Box_{\mathrm{T}}\right] v_z = v_z^0, \tag{35}$$

$$\left[1 + \ell_1^2 \Box_{\mathrm{T}}\right] \beta_{zx} = \beta_{zx}^0, \tag{36}$$

$$\left[1 + \ell_1^2 \Box_{\mathrm{T}}\right] \beta_{zy} = \beta_{zy}^0, \tag{37}$$

with the following $(1+2)$-dimensional d'Alembert operator (wave operator)

$$\Box_{\mathrm{T}} = \frac{1}{c_{\mathrm{T}}^2} \partial_{tt} - \Delta. \tag{38}$$

In addition, using (1) we may derive from (35)–(37) inhomogeneous Klein–Gordon equations for the non-vanishing components T_{zxy}, I_{zx}, I_{zy}

$$\left[1 + \ell_1^2 \Box_{\mathrm{T}}\right] T_{zxy} = T_{zxy}^0, \tag{39}$$

$$\left[1 + \ell_1^2 \Box_{\mathrm{T}}\right] I_{zx} = I_{zx}^0, \tag{40}$$

$$\left[1 + \ell_1^2 \Box_{\mathrm{T}}\right] I_{zy} = I_{zy}^0. \tag{41}$$

In field theory, Klein–Gordon equations describe massive fields (see, e.g., [27]). That means that a dislocation is a massive gauge field.

From the condition (34), we find for the inertia term of a screw dislocation

$$d_2 + d_3 = \frac{c_1}{c_{\mathrm{T}}^2} = \rho \ell_1^2, \tag{42}$$

that it is given in terms of the characteristic length scale ℓ_1.

4. Nonuniformly moving screw dislocation

Now we consider a nonuniformly moving screw dislocation at the position $(\xi(t), \eta(t))$ at time t. The dislocation line and the Burgers vector $b = b_z$ are parallel to the z-axis. The dislocation velocity has components: $V_x = \dot{\xi}(t)$, $V_y = \dot{\eta}(t)$.

At first, we want to find the gauge theoretical solutions of the dislocation density and the dislocation currents for a nonuniformly moving screw dislocation. Thus, we have to solve Eqs. (39)–(41), where the right-hand sides are given by the following sources

$$T_{zxy}^0 = b\delta\big(x - \xi(t)\big)\delta\big(y - \eta(t)\big), \tag{43}$$

$$I_{zx}^0 = V_y T_{zxy}^0, \tag{44}$$

$$I_{zy}^0 = -V_x T_{zxy}^0. \tag{45}$$

We consider the situation where the source terms have acted after initial quiescence at $t \to -\infty$. The solutions for the dislocation density and the dislocation currents are the convolution integrals[3]

$$T_{zxy} = G^{\mathrm{KG}} * T_{zxy}^0, \tag{46}$$

$$I_{zx} = G^{\mathrm{KG}} * I_{zx}^0, \tag{47}$$

$$I_{zy} = G^{\mathrm{KG}} * I_{zy}^0, \tag{48}$$

where the Green function of the $(1+2)$-dimensional Klein–Gordon equation is defined by

[3] The convolution is defined by: $T_{zxy}(\boldsymbol{x}, t) = G^{\mathrm{KG}} * T_{zxy}^0 = \int_{-\infty}^{t} \mathrm{d}t' \int \mathrm{d}\boldsymbol{x}' G^{\mathrm{KB}}(\boldsymbol{x}, t, \boldsymbol{x}', t') T_{zxy}^0(\boldsymbol{x}', t')$.

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]G^{\mathrm{KG}}=\delta(t)\delta(x)\delta(y). \tag{49}$$

The Green function of the Klein–Gordon equation in $(1+2)$ dimensions is given by [28]

$$G^{\mathrm{KG}}=\frac{1}{2\pi\ell_1^2}\frac{H(t-r/c_{\mathrm{T}})}{\sqrt{t^2-r^2/c_{\mathrm{T}}^2}}\cos\left(\frac{\sqrt{c_{\mathrm{T}}^2t^2-r^2}}{\ell_1}\right),\quad r^2=x^2+y^2, \tag{50}$$

where H is the Heaviside step function. Substituting (43)–(45) and (50) into (46)–(48), the spatial integration can be performed to give the results

$$T_{zxy}=\frac{b}{2\pi\ell_1^2}\int_{-\infty}^{t_{\mathrm{T}}}\mathrm{d}t'\,\frac{1}{S_{\mathrm{T}}}\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right), \tag{51}$$

$$I_{zx}=\frac{b}{2\pi\ell_1^2}\int_{-\infty}^{t_{\mathrm{T}}}\mathrm{d}t'\,\frac{V_y(t')}{S_{\mathrm{T}}}\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right), \tag{52}$$

$$I_{zy}=-\frac{b}{2\pi\ell_1^2}\int_{-\infty}^{t_{\mathrm{T}}}\mathrm{d}t'\,\frac{V_x(t')}{S_{\mathrm{T}}}\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right) \tag{53}$$

with the notations

$$\bar{x}=x-\xi(t'),\qquad \bar{y}=y-\eta(t'),\qquad \bar{t}=t-t',\qquad \bar{R}^2=\bar{x}^2+\bar{y}^2,$$
$$S_{\mathrm{T}}^2=\bar{t}^2-\frac{\bar{R}^2}{c_{\mathrm{T}}^2},\qquad t_{\mathrm{T}}=t-\frac{\bar{R}}{c_{\mathrm{T}}}. \tag{54}$$

Note that t_{T} is called retarded time, which is the root of the equation $S_{\mathrm{T}}^2=0$ and is less than t, and $(\bar{x},\bar{y})$ is the distance between the field point (x,y) and the position of the dislocation (ξ,η). Here, ξ and η are the positions in the x and y directions of the dislocation at time t', V_x and V_y are the velocity components of the dislocation at the same time t'. The retarded time t_{T} is the time before t, when the dislocation caused an excitation of the dislocation field, which moves from (ξ,η) to (x,y) in the time $\bar{R}/c_{\mathrm{T}}$. In addition, $t-t_{\mathrm{T}}=\bar{R}/c_{\mathrm{T}}$ is the time when the dislocation fields move from (ξ,η) to (x,y) with velocity c_{T}. Thus, from each point $\boldsymbol{x}'$, the dislocation fields (51)–(53) draw contributions emitted at all times t' from $-\infty$ up to t_{T}. That is the reason why Eshelby [5] said: 'The dislocation is haunted by its past'.

4.1. Solutions for the elastic fields in terms of potential functions

If we multiply Eqs. (35)–(37) with $\Box_{\mathrm{T}}$ and use the 'classical' result [6,3], we obtain

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}v_z=I^0_{zx,x}+I^0_{zy,y}, \tag{55}$$

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}\beta_{zx}=T^0_{zxy,y}+\frac{1}{c_{\mathrm{T}}^2}\dot{I}^0_{zx}, \tag{56}$$

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}\beta_{zy}=T^0_{zyx,x}+\frac{1}{c_{\mathrm{T}}^2}\dot{I}^0_{zy}, \tag{57}$$

as a set of fourth-order partial differential equations. Eqs. (55)–(57) have the two-dimensional form of the Bopp–Podolsky equations [29,30] (see also [28]) in generalized electrodynamics, introduced by Bopp and Podolsky in order to avoid singularities in electrodynamics. As source terms only the classical dislocation density T^0_{zxy} and dislocation currents I^0_{zx}, I^0_{zy} are acting.

Following Eshelby [6], we express the elastic velocity and the elastic distortion in terms of 'potential' functions $(F,\ A_x,\ A_y)$:

$$v_z=A_{y,x}-A_{x,y}, \tag{58}$$

$$\beta_{zx}=F_{,y}+\frac{1}{c_{\mathrm{T}}^2}\dot{A}_y, \tag{59}$$

$$\beta_{zy}=-F_{,x}-\frac{1}{c_{\mathrm{T}}^2}\dot{A}_x. \tag{60}$$

Then, if A_x, A_y and F are chosen to satisfy the subsidiary condition

$$A_{x,x}+A_{y,y}+\dot{F}=0, \tag{61}$$

the field equations (55)–(57) become equivalent to the following inhomogeneous Bopp–Podolsky equations

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}F=T^0_{zxy}, \tag{62}$$

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}A_x=-I^0_{zy}, \tag{63}$$

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}A_y=I^0_{zx}. \tag{64}$$

The solution of these equations for an infinite medium is defined by

$$F = G^{\mathrm{BP}} * T^0_{zxy}, \tag{65}$$

$$A_x = -G^{\mathrm{BP}} * I^0_{zy}, \tag{66}$$

$$A_y = G^{\mathrm{BP}} * I^0_{zx}, \tag{67}$$

where G^{BP} is the Green function of the Bopp–Podolsky equation (or wave-Klein–Gordon equation)

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}G^{\mathrm{BP}} = \delta(t)\delta(x)\delta(y). \tag{68}$$

As in generalized electrodynamics, we solve (68) with the help of two fields

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]G^{\mathrm{BP}} = G^{\Box}, \qquad \Box_{\mathrm{T}}G^{\Box} = \delta(t)\delta(x)\delta(y) \tag{69}$$

and we get

$$G^{\mathrm{BP}} = G^{\Box} - \ell_1^2 G^{\mathrm{KG}}, \tag{70}$$

where the first field is the Green function of the two-dimensional wave equation [7,31]

$$G^{\Box} = \frac{1}{2\pi}\frac{H(t-r/c_{\mathrm{T}})}{\sqrt{t^2-r^2/c_{\mathrm{T}}^2}} \tag{71}$$

and the second one is the Green function of the Klein–Gordon equation (50). Finally, we obtain for the Green function of the $(1+2)$-dimensional Bopp–Podolsky equation

$$G^{\mathrm{BP}} = \frac{1}{2\pi}\frac{H(t-r/c_{\mathrm{T}})}{\sqrt{t^2-r^2/c_{\mathrm{T}}^2}}\left[1-\cos\left(\frac{\sqrt{c_{\mathrm{T}}^2t^2-r^2}}{\ell_1}\right)\right]. \tag{72}$$

Substituting from (43)–(45) for the case of a single dislocation, the spatial integrations can be performed to give the results for the potential functions of a screw dislocation

$$F = \frac{b}{2\pi}\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{1}{S_{\mathrm{T}}}\left[1-\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right)\right], \tag{73}$$

$$A_x = \frac{b}{2\pi}\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{V_x(t')}{S_{\mathrm{T}}}\left[1-\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right)\right], \tag{74}$$

$$A_y = \frac{b}{2\pi}\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{V_y(t')}{S_{\mathrm{T}}}\left[1-\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right)\right]. \tag{75}$$

In analogy to electromagnetic field theory, we can call (73)–(75) the (two-dimensional) 'Liénard–Wiechert potentials' of a screw dislocation in the gauge theory of dislocations.

4.2. Radiation part of the elastic fields

On the other hand, using (44) and (45), we want to solve directly Eqs. (55)–(57). Consequently, we have to solve the equations

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}v_z = V_yT^0_{zxy,x} - V_xT^0_{zxy,y}, \tag{76}$$

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}\beta_{zx} = \frac{\dot V_y}{c_{\mathrm{T}}^2}T^0_{zxy} + \left(1-\frac{V_y^2}{c_{\mathrm{T}}^2}\right)T^0_{zxy,y} - \frac{V_xV_y}{c_{\mathrm{T}}^2}T^0_{zxy,x}, \tag{77}$$

$$\left[1+\ell_1^2\Box_{\mathrm{T}}\right]\Box_{\mathrm{T}}\beta_{zy} = -\frac{\dot V_x}{c_{\mathrm{T}}^2}T^0_{zxy} - \left(1-\frac{V_x^2}{c_{\mathrm{T}}^2}\right)T^0_{zxy,x} + \frac{V_xV_y}{c_{\mathrm{T}}^2}T^0_{zxy,y}, \tag{78}$$

where the right-hand side is given by (43). It can be seen that only the elastic distortions contain acceleration parts. The elastic velocity does not depend on dislocation acceleration. Using the Green function (72), we obtain the result for the elastic distortions

$$\beta_{zx} = \frac{b}{2\pi c_{\mathrm{T}}^2}\left[\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{\dot V_y(t')}{S_{\mathrm{T}}}\left[1-\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right)\right] + \frac{\partial}{\partial y}\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{c_{\mathrm{T}}^2-V_y^2(t')}{S_{\mathrm{T}}}\left[1-\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right)\right]\right.$$
$$\left. - \frac{\partial}{\partial x}\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{V_x(t')V_y(t')}{S_{\mathrm{T}}}\left[1-\cos\left(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\right)\right]\right], \tag{79}$$

$$\beta_{zy} = -\frac{b}{2\pi c_{\mathrm{T}}^2}\Bigg[\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{\dot{V}_x(t')}{S_{\mathrm{T}}}\Bigg[1-\cos\Bigg(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\Bigg)\Bigg] + \frac{\partial}{\partial x}\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{c_{\mathrm{T}}^2 - V_x^2(t')}{S_{\mathrm{T}}}\Bigg[1-\cos\Bigg(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\Bigg)\Bigg]$$
$$- \frac{\partial}{\partial y}\int\limits_{-\infty}^{t_{\mathrm{T}}} \mathrm{d}t'\,\frac{V_x(t')V_y(t')}{S_{\mathrm{T}}}\Bigg[1-\cos\Bigg(\frac{c_{\mathrm{T}}S_{\mathrm{T}}}{\ell_1}\Bigg)\Bigg]\Bigg]. \tag{80}$$

The solution of Eq. (76) for the elastic velocity has the same expression as given in Eq. (58) with (74) and (75). It can be seen that the fields given by (79) and (80) consist of two parts:

(a) Field depending on the dislocation velocities V_x and V_y alone and proportional to spatial derivatives – velocity field.
(b) Field depending on the dislocation accelerations $\dot{V}_x$ and $\dot{V}_y$ and proportional to $1/S_{\mathrm{T}}$ – acceleration field or radiation field.

All our results depend on ℓ_1 which is a characteristic length scale of the gauge theory of dislocations.

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